EAS 410

Practical 1:
Serial Communication and Computer Memory

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EAS 410 Practical Objectives

- Implement computer hardware modules on an FPGA to provide a practical understanding of computer architecture

- Modules that will be implemented:
  - Practical 1: UART
  - Practical 2: RAM
  - Practical 3: CPU

Practical 1 Overview

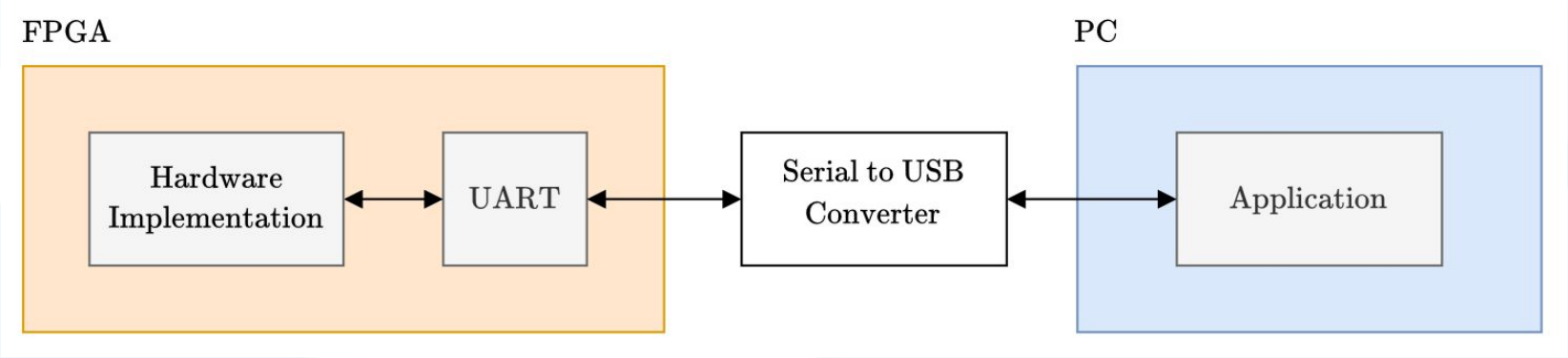
Task A: UART Implementation

Task B: Memory Design

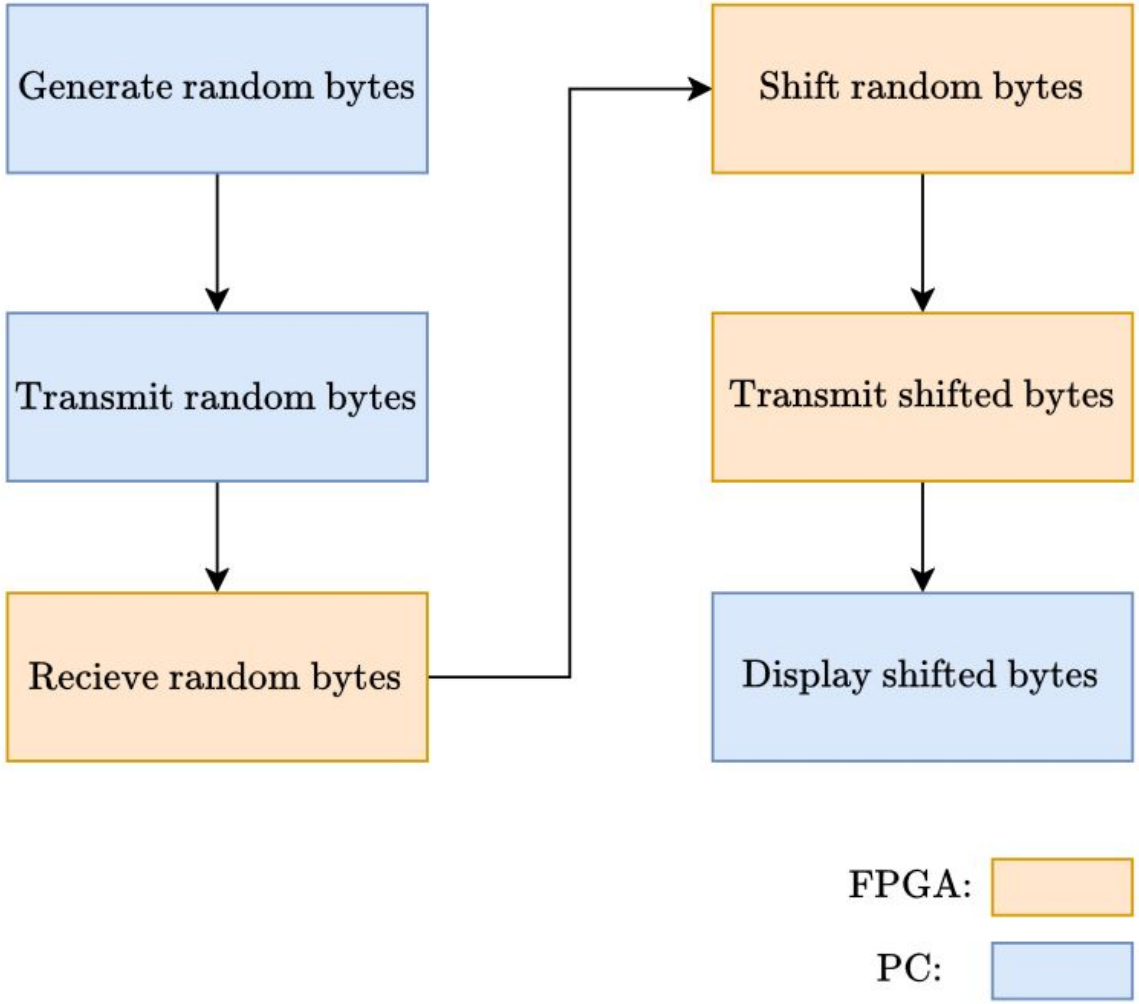
Task A: UART

Overview

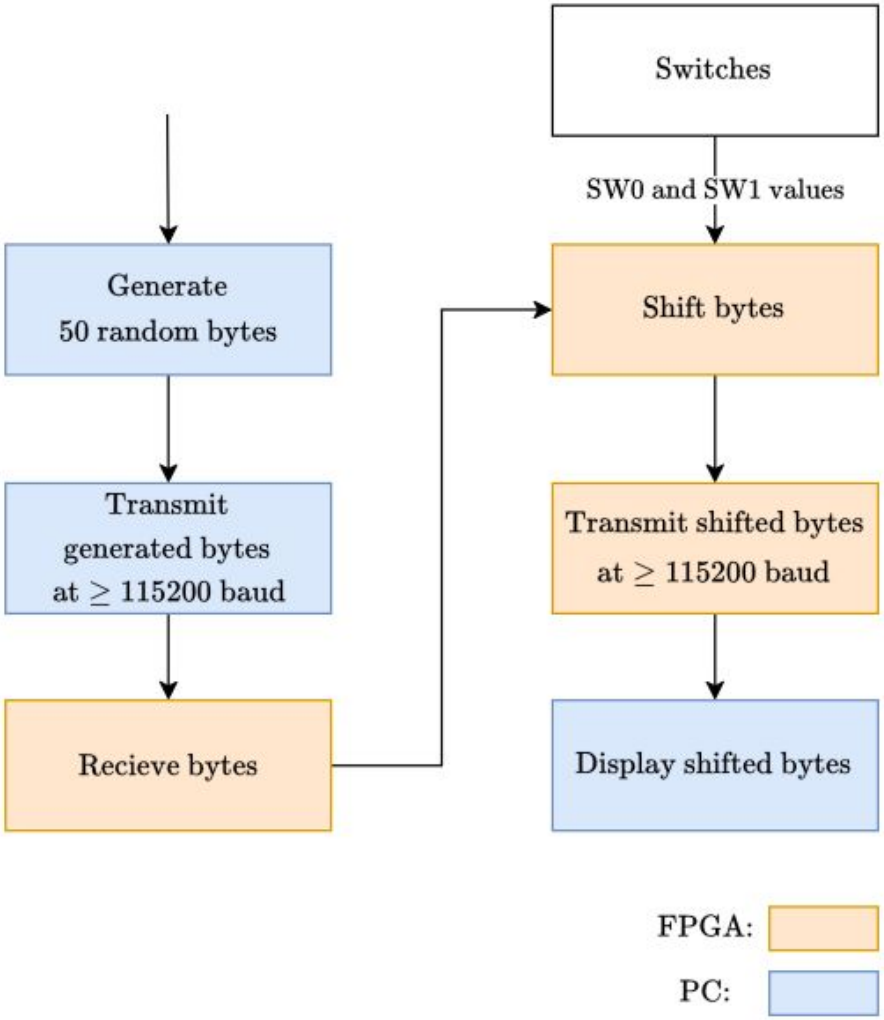
- UART allows communication with FPGA to control implemented hardware via an application running on a PC
- PC application is the user interface of the system



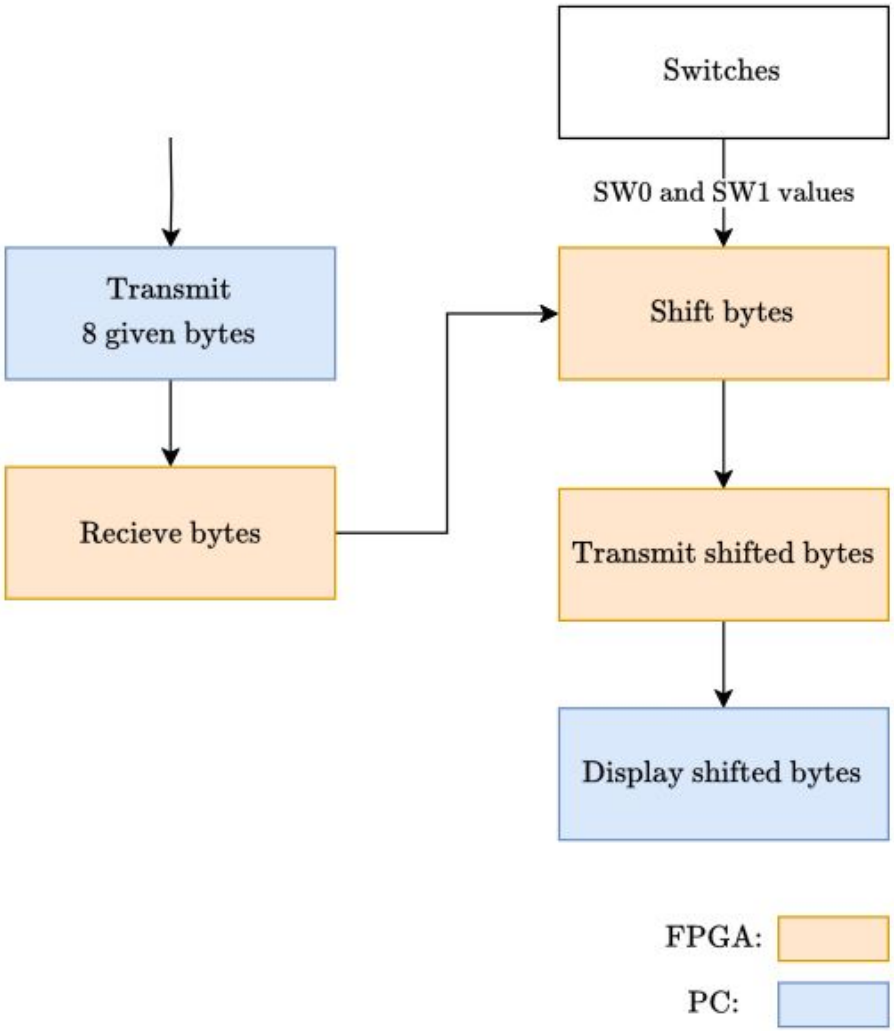
Task A: UART


Generate random bytes
Transmit random bytes
Recieve random bytes
Shift random bytes
Transmit shifted bytes
Display shifted bytes
FPGA:
PC:

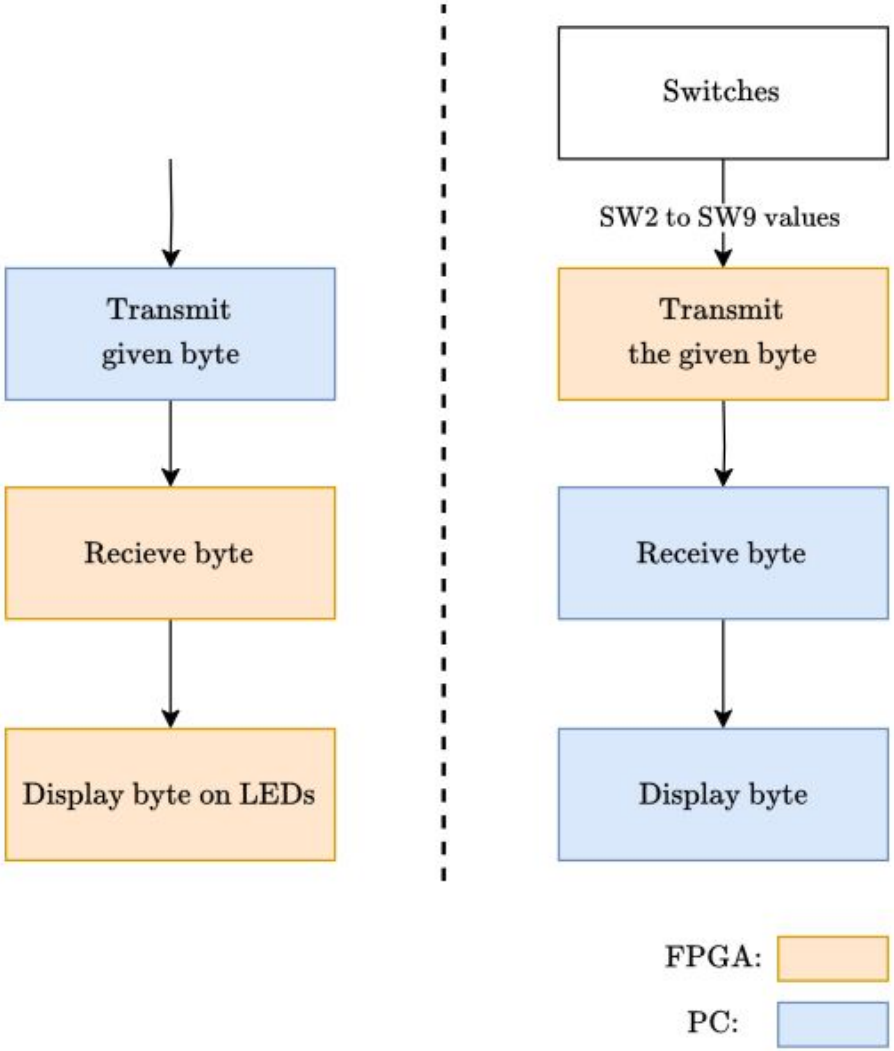
Task A: Approach 1 (Max 100%)



Task A: Approach 2 (Max 75%)



Task A: Approach 3 (Max 50%)



Task A: Hardware Implementation

SW0	SW1	Action
0	0	One left shift.
0	1	One right shift.
1	0	Two left shifts.
1	1	Two right shifts.

One left shift: 1101 0000 becomes 1010 0001 (not required for approach 3)

Task A: Software Implementation

```
1 0000 0000 - 0000 0000
2 0101 0101 - 1010 1010
3 1111 0000 - 1110 0001
4 1100 1100 - 1001 1000 Error!
                .
                .
                .
47 0001 1111 - 0011 1110
48 0110 1010 - 1101 1000
49 0001 1000 - 0011 0000
50 0100 0011 - 1000 0110

Errors: 1
Transmit Time: 4 ms = 50 x 8 bits / 0.004 s = 100 000 bps
Receive Time: 5 ms = 50 x 8 bits / 0.005 s = 80 000 bps
```

Display errors - compare received with expected byte after shift and calculate the approximate baud rate (not required for approach 3)

Task A: Serial Communication

FPGA:

- May use Verilog or VHDL

PC Application:

- Any language allowed (C++, Python etc.)

Any method of converting USB to Serial may be used – off-the-shelf converter, Arduino Uno etc.

Task B: Memory Design

- 8-bit, 256 byte RAM.
- Design using 8x8x1 bit chips.
- Memory access:
  - Nibble (4 bits)
  - Word (8 bits)
- Read and write capabilities.
- Will be implemented in practical 2, only design for practical 1

Task B: Memory Design

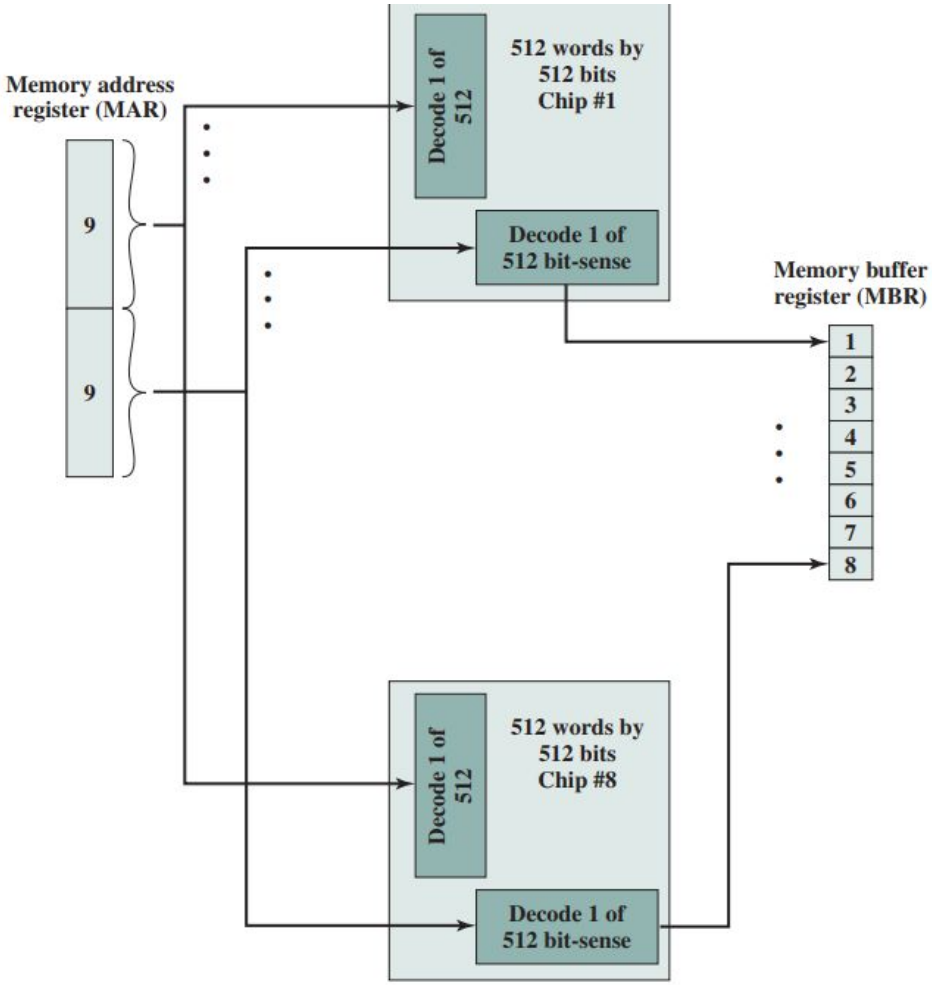


Figure 5.5 256-KByte Memory Organization

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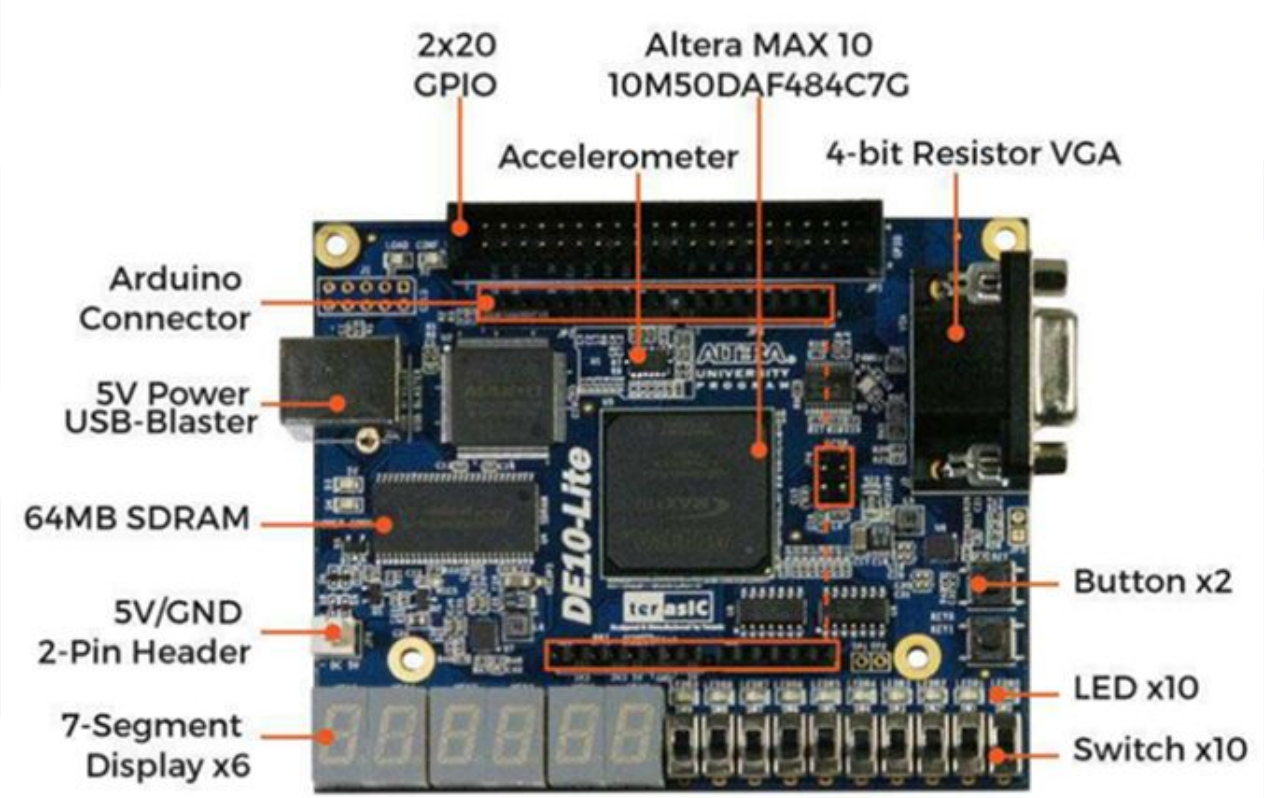
FPGA Board

- Still have the old FPGA and don't have a DE10-Lite?
- Breakout extra I/O if required



Groups

- Sign-up on the AMS under Demo 1 by 20 February
- 2 students per group
- Groups will be auto assigned after the deadline.
- Clashes with other modules?

Demonstration

- Date: Wednesday, 4 March 2026
- Venue: Netlabs
- 15 minutes per demo
- Group demo slots will be posted the day before the practical
- Hardware must be ready
- Be prepared to show your code and lab books
- Students are expected to have knowledge of all aspects of the practical assignment

Summary of Findings (Reports)

- Consult practical guide for required content.
- Used to evaluate understanding of the practical, and your engineering process.
- Use figures!
- Figures should be vector graphics (pdf / svg).

Functional Simulation

- Required for everything implemented on the FPGA.
- Any method allowed: ModelSim, Quartus etc.
- Keep their limits in mind. What are they?

Additional Information

- Practical mark weighting 50/50 between demo and lab book/questions.
- Must meet sub-minimum requirements.
- Must obtain a mark of 50% or above for both the demo and the report.

Ask questions on Telegram or email up.eas410@gmail.com

Questions?